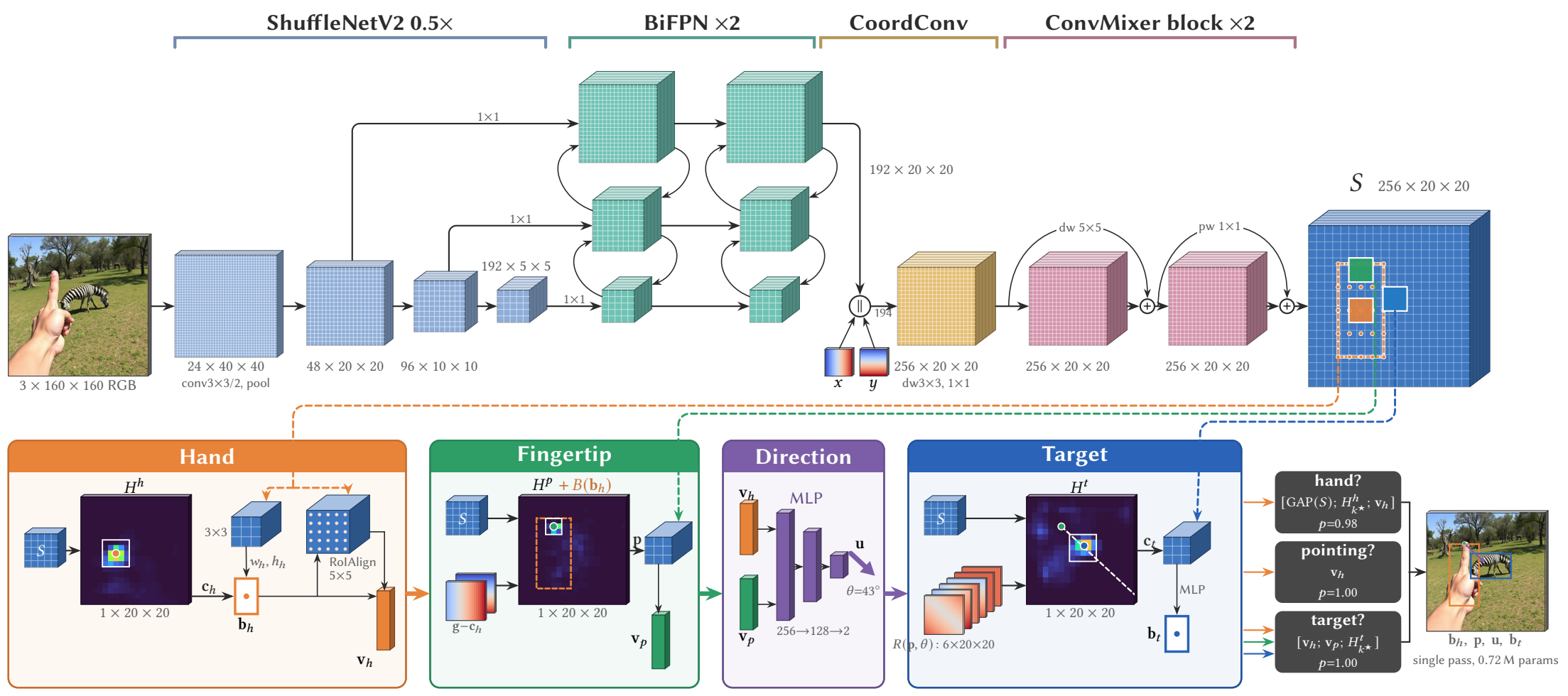

ShuffleNetV2 0.5×
BiFPN ×2
CoordConv
ConvMixer block ×2
3 × 160 × 160 RGB
24 × 40 × 40
conv3×3/2, pool
48 × 20 × 20
96 × 10 × 10
192 × 5 × 5
1×1
1×1
1×1
192 × 20 × 20
194
256 × 20 × 20
dw3×3, 1×1
x
y
256 × 20 × 20
dw 5×5
256 × 20 × 20
pw 1×1
S
256 × 20 × 20
Hand
H^h
S
1 × 20 × 20
3×3
w_h, h_h
c_h
b_h
RoIAlign
5×5
v_h
Fingertip
H^p + B(b_h)
S
g−c_h
1 × 20 × 20
p
v_p
Direction
v_h
MLP
u
θ=43°
v_p
256→128→2
Target
H^t
S
1 × 20 × 20
c_t
MLP
b_t
R(p, θ) : 6×20×20
hand?
[GAP(S); H^h_k⋆ ; v_h]
p=0.98
pointing?
v_h
p=1.00
target?
[v_h; v_p; H^t_k⋆]
p=1.00
b_h, p, u, b_t
single pass, 0.72 M params